

Amazon S3 Versioning Lab

To Begin with the Lab

Summary of the Lab

In this lab, **Amazon S3 Versioning** was demonstrated by creating buckets with versioning turned off and then turned on. First, the same file was uploaded twice to a bucket with versioning disabled, which caused the newer file to overwrite the older one. Then, a second bucket with versioning enabled was used so the same file could keep multiple versions, including the original version and the updated current version. The lab also showed that deleting an object in a versioned bucket creates a delete marker instead of permanently removing the file, which makes recovery possible. Finally, it was explained that versioning can be suspended but not fully disabled once enabled, and that extra stored versions can increase storage cost.

Understanding Amazon S3 Versioning

Amazon S3 Versioning lets you keep multiple versions of the same object inside one bucket. It exists so you can protect data from accidental overwrite or accidental deletion. Without versioning, uploading a file with the same name replaces the old file, and deleting it removes it permanently. With versioning enabled, S3 keeps the older object as a previous version and adds a delete marker when the file is removed, which means the file can still be recovered later.

Example:

A file named file1.txt is uploaded with the content hello. Later, the same file name is uploaded again with the content word. With versioning disabled, the original content is overwritten. With versioning enabled, S3 stores both versions, and the old file can still be restored by removing the delete marker or selecting the earlier version.

Lab Steps

Step 1 – Create a bucket with versioning disabled

- Sign in to the **AWS Management Console**.
- Open **Amazon S3**.
- Click **Create bucket**.
- Enter the bucket name **s3-version-disabled**.

Bucket type [Info](#)

☒ **General purpose**
Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.

☐ **Directory**
Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

Bucket namespace
Choose the namespace where you want to create your bucket. [Learn more](#)

☒ **Global namespace**
By default, S3 creates general purpose buckets in the global namespace.

☐ **Account Regional namespace (recommended)**
General purpose buckets created in your account Regional namespace are unique to your account. These buckets can never be created by another AWS account.

Bucket name [Info](#)

Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). [Learn more](#)

Copy settings from existing bucket - optional
Only the bucket settings in the following configuration are copied.

[Choose bucket](#)

Format: s3://bucket/prefix

Object Ownership [Info](#)
Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

Object Ownership

☒ **ACLs disabled (recommended)**
All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.

☐ **ACLs enabled**
Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.

- Keep bucket versioning disabled.

Block Public Access settings for this bucket
Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to this bucket and its objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to this bucket or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

☒ **Block all public access**
Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.

☐ **Block public access to buckets and objects granted through new access control lists (ACLs)**
S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.

☐ **Block public access to buckets and objects granted through any access control lists (ACLs)**
S3 will ignore all ACLs that grant public access to buckets and objects.

☐ **Block public access to buckets and objects granted through new public bucket or access point policies**
S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.

☐ **Block public and cross-account access to buckets and objects through any public bucket or access point policies**
S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

Bucket Versioning
Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures. [Learn more](#)

Bucket Versioning

☒ **Disable**

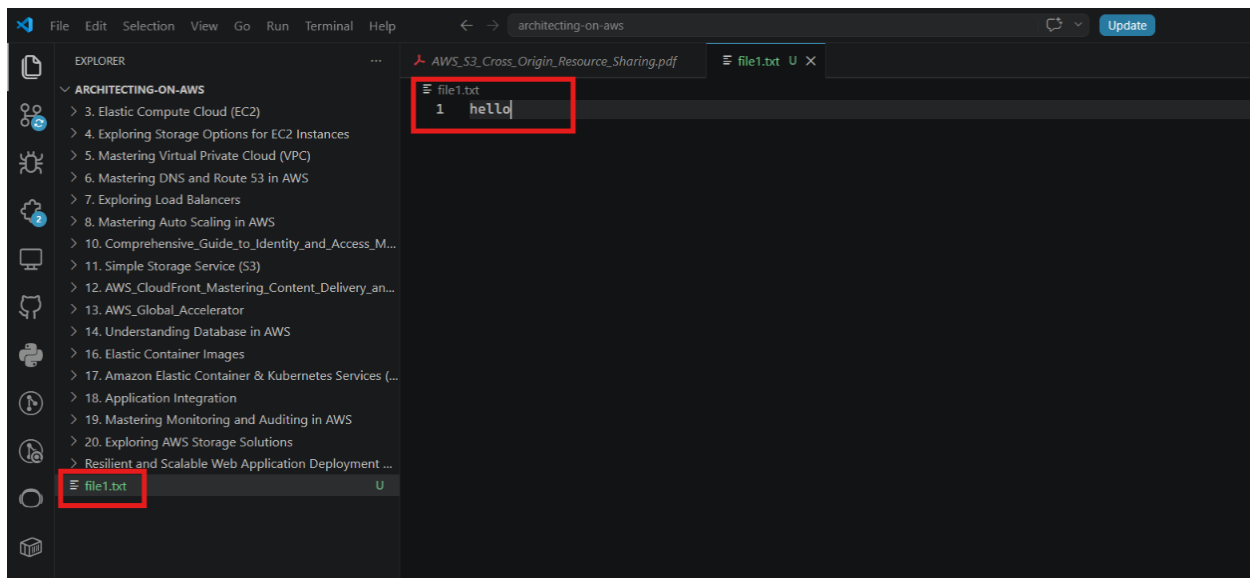
☐ **Enable**

Tags - optional
You can use bucket tags to analyze, manage and specify permissions for a bucket. [Learn more](#)

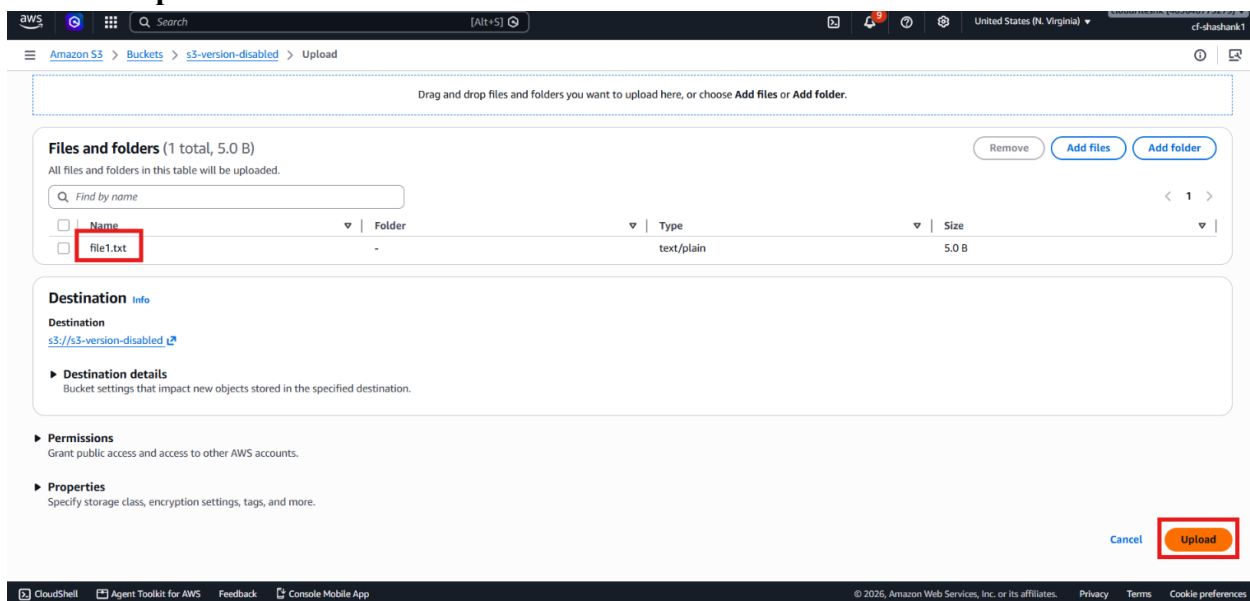
- Create the bucket.

Step 2 – Upload the first object

- Create a local text file named file1.txt.
- Write the content hello.



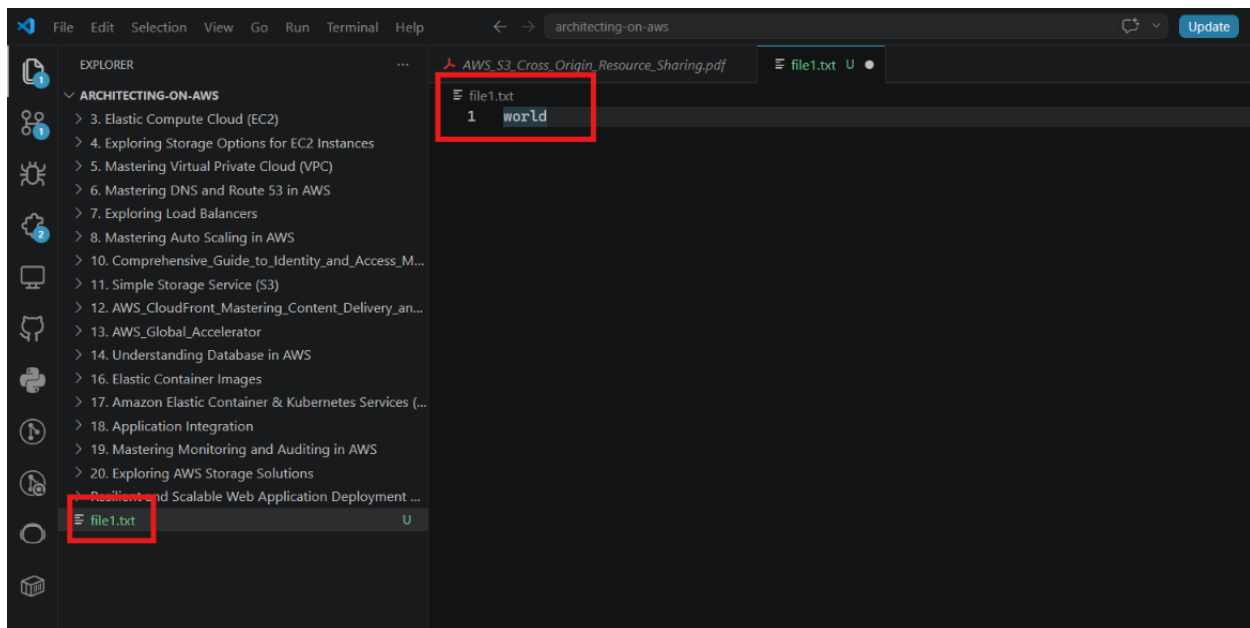
- Open the bucket.
- Click **Upload**.



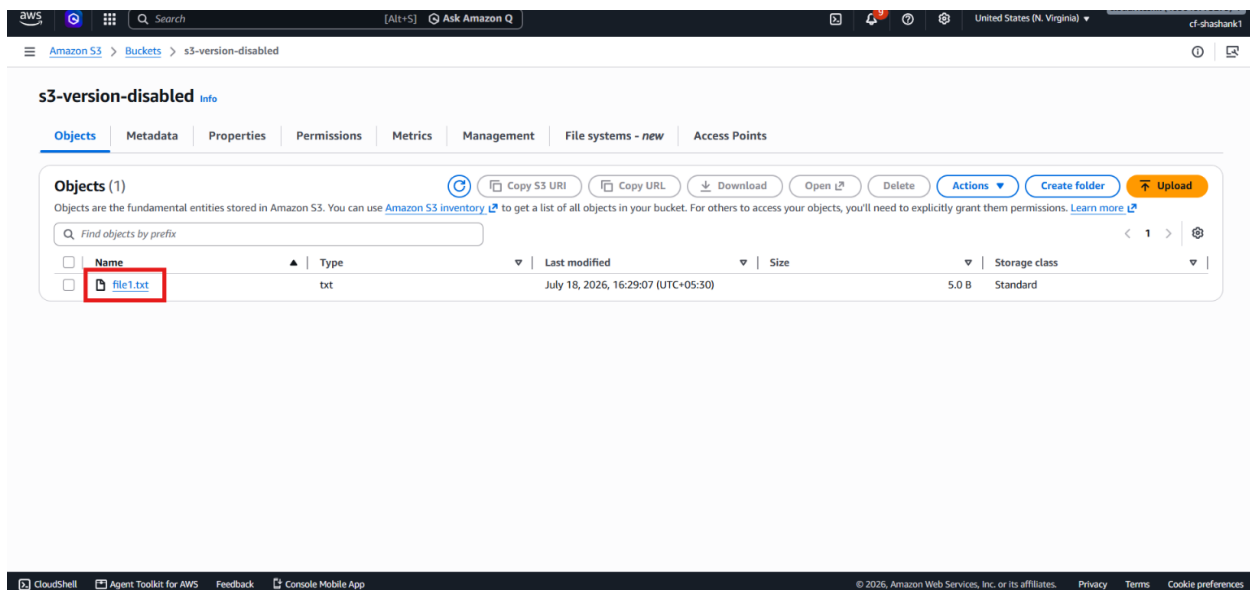
- Add file1.txt.
- Click **Upload** to store the file in the bucket.

Step 3 – Upload the same object again

- Edit the same local file.
- Change the content to world.

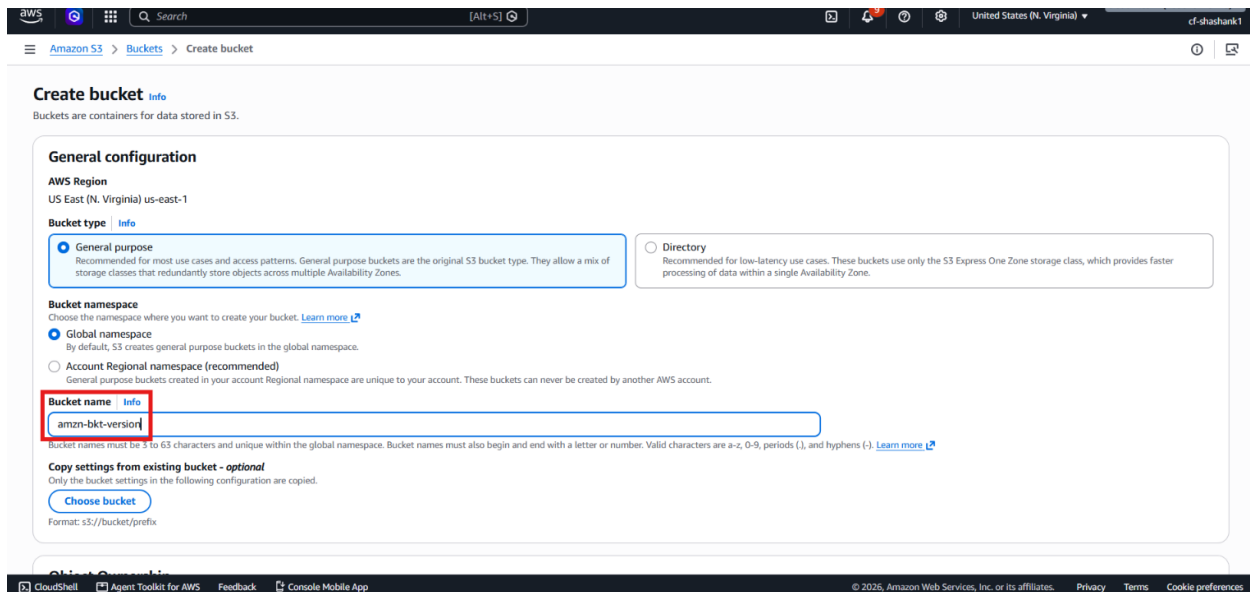


- Upload the file again with the same name file1.txt.
- Open the object and verify that the new content replaced the old one because versioning is disabled.



Step 4 – Create a bucket with versioning enabled

- Create a new bucket named **amzn-bkt-version**.



Create bucket [info](#)

Buckets are containers for data stored in S3.

General configuration

AWS Region
US East (N. Virginia) us-east-1

Bucket type [info](#)

☒ **General purpose**
Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.

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Bucket name [info](#)

amzn-bkt-version

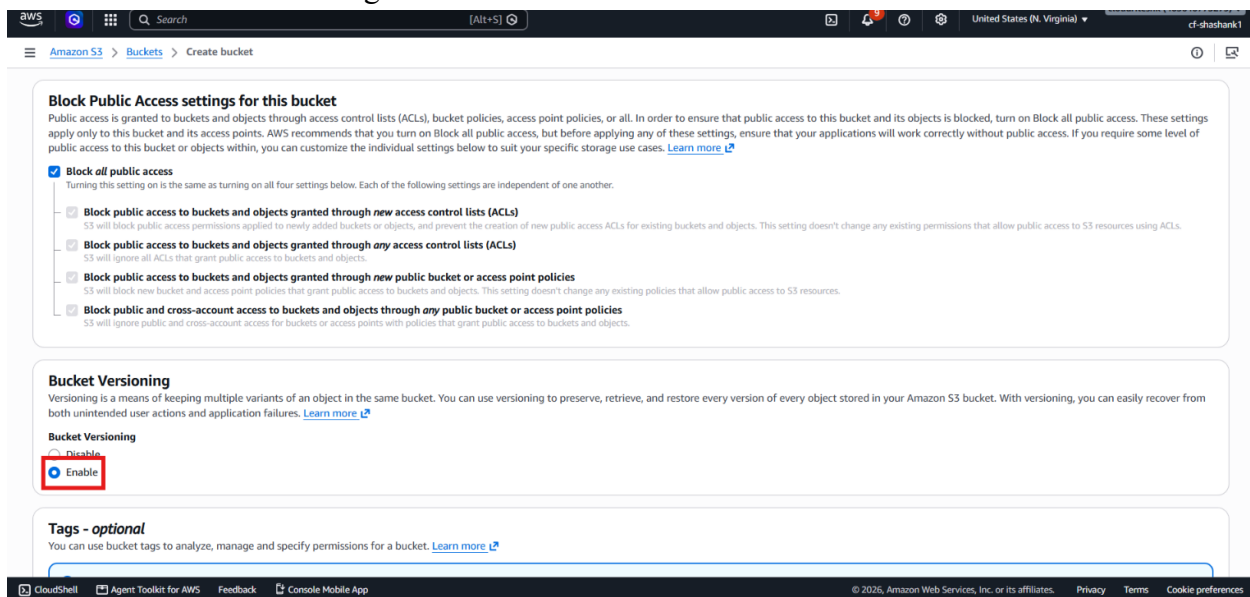
Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). [Learn more](#)

Copy settings from existing bucket - optional
Only the bucket settings in the following configuration are copied.

[Choose bucket](#)

Format: s3://bucket/prefix

- Open the bucket **Properties**.
- Enable bucket versioning.



Block Public Access settings for this bucket

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Bucket Versioning

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Bucket Versioning

☒ **Enable**

Tags - optional
You can use bucket tags to analyze, manage and specify permissions for a bucket. [Learn more](#)

- Create the bucket.

Step 5 – Upload the file again with versioning enabled

- Upload file1.txt with the content hello.

The screenshot shows the AWS Management Console interface for uploading a file. The breadcrumb navigation indicates the path: Amazon S3 > Buckets > amzn-bkt-version > Upload. A large dashed box at the top contains the instruction: "Drag and drop files and folders you want to upload here, or choose Add files or Add folder." Below this, a section titled "Files and folders (1 total, 5.0 B)" states "All files and folders in this table will be uploaded." It includes a search bar "Find by name" and a table with columns: Name, Folder, Type, and Size. The table contains one entry: "file1.txt" with a folder of "-", type of "text/plain", and size of "5.0 B". The "file1.txt" text is highlighted with a red box. Below the table is the "Destination" section, which includes a link to the bucket's S3 URL and a "Destination details" section. At the bottom right, there are "Cancel" and "Upload" buttons, with the "Upload" button highlighted by a red box.

Amazon S3 > Buckets > amzn-bkt-version > Upload

Drag and drop files and folders you want to upload here, or choose [Add files](#) or [Add folder](#).

Files and folders (1 total, 5.0 B)
All files and folders in this table will be uploaded.

Find by name

Name	Folder	Type	Size
file1.txt	-	text/plain	5.0 B

Destination [info](#)
Destination
[s3://amzn-bkt-version](#)

Destination details
Bucket settings that impact new objects stored in the specified destination.

Permissions
Grant public access and access to other AWS accounts.

Properties
Specify storage class, encryption settings, tags, and more.

[Cancel](#) [Upload](#)

- Upload the same file again after changing the content to world.
- Open the object and use the version toggle.

The screenshot shows the AWS Management Console interface for the 'amzn-bkt-version' bucket. The breadcrumb navigation is: Amazon S3 > Buckets > amzn-bkt-version. The page title is 'amzn-bkt-version' with an 'info' link. Below the title are tabs for 'Objects', 'Metadata', 'Properties', 'Permissions', 'Metrics', 'Management', 'File systems - new', and 'Access Points'. The 'Objects' tab is selected. It shows a list of objects with columns: Name, Type, Last modified, Size, and Storage class. The 'Show versions' toggle is turned on and highlighted with a red box. The table contains one object: 'file1.txt' with a type of 'txt', last modified on 'July 18, 2026, 16:48:44 (UTC+05:30)', size of '5.0 B', and storage class of 'Standard'. The 'file1.txt' text is highlighted with a red box.

Amazon S3 > Buckets > amzn-bkt-version

amzn-bkt-version [info](#)

[Objects](#) [Metadata](#) [Properties](#) [Permissions](#) [Metrics](#) [Management](#) [File systems - new](#) [Access Points](#)

Objects (1)
Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

☒ Show versions

Name	Type	Last modified	Size	Storage class
file1.txt	txt	July 18, 2026, 16:48:44 (UTC+05:30)	5.0 B	Standard

- Confirm that both the current version and the previous version are visible.

The screenshot shows the AWS Management Console for the bucket 'amzn-bkt-version'. The 'Objects' tab is selected, and the 'Show versions' button is highlighted. The table below shows two versions of the object 'file1.txt'.

	Name	Type	Version ID	Last modified	Size	Storage class
☐	file1.txt	txt	xZtdBW6KvEr2xI94sJWaoKdyYUGITOVZ	July 18, 2026, 16:48:44 (UTC+05:30)	5.0 B	Standard
☐	file1.txt	txt	ASAA1jRvcFBKqnx0e2aqIcMlqFUMzc5	July 18, 2026, 16:47:16 (UTC+05:30)	5.0 B	Standard

Step 6 – Test delete behavior without versioning

- Go back to the bucket where versioning is disabled.
- Delete file1.txt.

The screenshot shows the AWS Management Console for the bucket 's3-version-disabled'. The 'Objects' tab is selected, and the 'Delete' button is highlighted. The table below shows one object 'file1.txt'.

	Name	Type	Last modified	Size	Storage class
☒	file1.txt	txt	July 18, 2026, 16:29:07 (UTC+05:30)	5.0 B	Standard

- Type **delete permanently** when prompted.

Delete objects [Info](#)

⚠️ If a folder is selected for deletion, all objects in the folder will be deleted, and any new objects added while the delete action is in progress might also be deleted. If an object is selected for deletion, any new objects with the same name that are uploaded before the delete action is completed will also be deleted.

- Deleting the specified objects can't be undone.

[Learn more](#)

Specified objects

Find objects by name

Name	Type	Last modified	Size
file1.txt ↗	txt	July 18, 2026, 16:29:07 (UTC+05:30)	5.0 B

Permanently delete objects?

To confirm deletion, type *permanently delete* in the text input field.

permanently delete

[Cancel](#) [Delete objects](#)

- Confirm that the object is removed permanently and cannot be restored.

s3-version-disabled [Info](#)

[Objects](#) [Metadata](#) [Properties](#) [Permissions](#) [Metrics](#) [Management](#) [File systems - new](#) [Access Points](#)

Objects (0)

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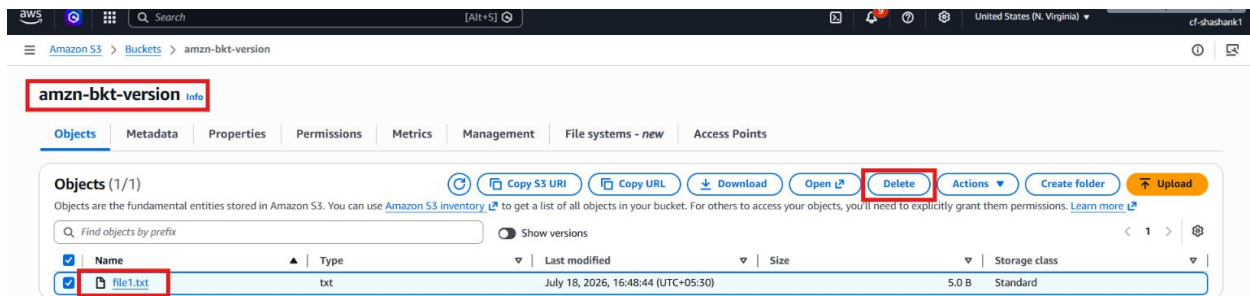
Find objects by prefix

Name	Type	Last modified	Size	Storage class
No objects You don't have any objects in this bucket.				

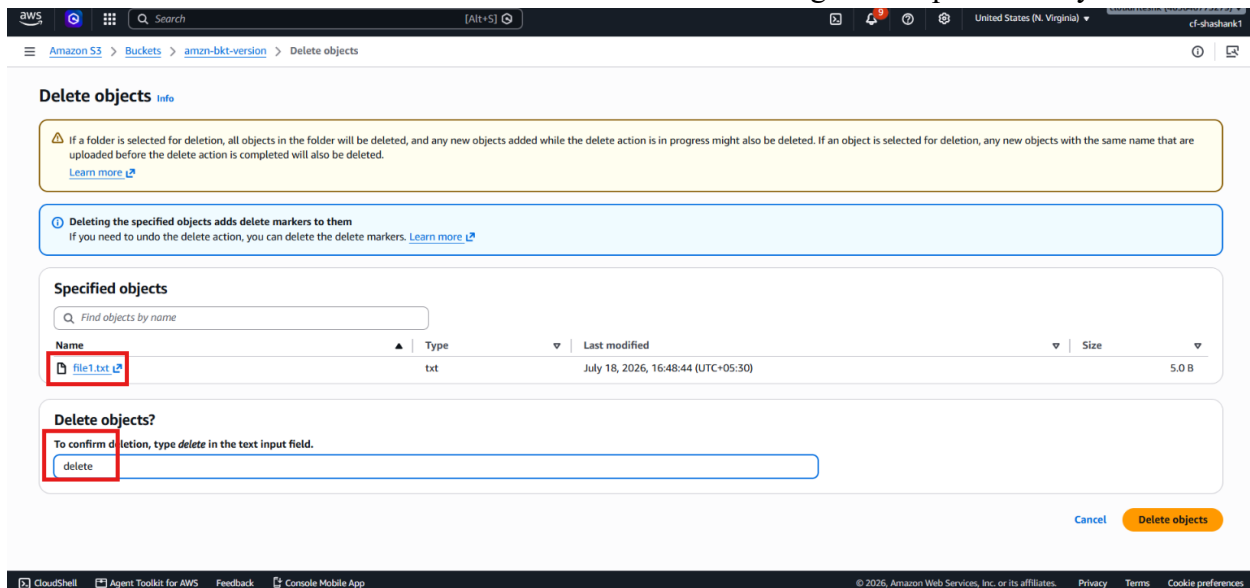
[Upload](#)

Step 7 – Test delete behavior with versioning enabled

- Open the versioned bucket.
- Delete file1.txt.



- Confirm that S3 creates a delete marker instead of removing the file permanently. =



- Open the version view and confirm that the original file still exists as a previous version.

Amazon S3 > Buckets > amzn-bkt-version

amzn-bkt-version

Objects (3)

Find objects by prefix

Show versions

	Name	Type	Version ID	Last modified	Size	Storage class
☐	file1.txt	Delete marker	x0ubpVGSXITezxrQoPSaqUSdGbKk5zJl	July 18, 2026, 16:54:30 (UTC+05:30)	0 B	-
☐	file1.txt	txt	xZTdBW6kVEr2xI94sJWaoKdyYUGITOVZ	July 18, 2026, 16:48:44 (UTC+05:30)	5.0 B	Standard
☐	file1.txt	txt	ASAA1jRvcFBKqnx0ezaqlcMlqFUMzc5	July 18, 2026, 16:47:16 (UTC+05:30)	5.0 B	Standard

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Step 8 – Restore the deleted file

- Select the delete marker version.

Amazon S3 > Buckets > amzn-bkt-version

amzn-bkt-version

Objects (1/3)

Find objects by prefix

Show versions

	Name	Type	Version ID	Last modified	Size	Storage class
☒	file1.txt	Delete marker	x0ubpVGSXITezxrQoPSaqUSdGbKk5zJl	July 18, 2026, 16:54:30 (UTC+05:30)	0 B	-
☐	file1.txt	txt	xZTdBW6kVEr2xI94sJWaoKdyYUGITOVZ	July 18, 2026, 16:48:44 (UTC+05:30)	5.0 B	Standard
☐	file1.txt	txt	ASAA1jRvcFBKqnx0ezaqlcMlqFUMzc5	July 18, 2026, 16:47:16 (UTC+05:30)	5.0 B	Standard

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- Delete the delete marker permanently.

aws [Search] [Alt+S] United States (N. Virginia) cf-shashank1

Amazon S3 > Buckets > amzn-bkt-version > Delete objects

Delete objects [Info](#)

⚠️

- If a folder is selected for deletion, all objects in the folder will be deleted, and any new objects added while the delete action is in progress might also be deleted. If an object is selected for deletion, any new objects with the same name that are uploaded before the delete action is completed will also be deleted.
- Deleting the specified objects can't be undone.

[Learn more](#)

Specified objects

Find objects by name

Name	Version ID	Type	Last modified	Size
file1.txt	x0ubpVGSX0TezxrQoPSaqUSdGbKkSzJl	Delete marker	July 18, 2026, 16:54:30 (UTC+05:30)	0 B

Permanently delete objects?

To confirm deletion, type *permanently delete* in the text input field.

permanently delete

Cancel **Delete objects**

- Switch off the version toggle.

aws [Search] [Alt+S] United States (N. Virginia) cf-shashank1

Amazon S3 > Buckets > amzn-bkt-version

amzn-bkt-version [Info](#)

Objects Metadata Properties Permissions Metrics Management File systems - new Access Points

Objects (2)

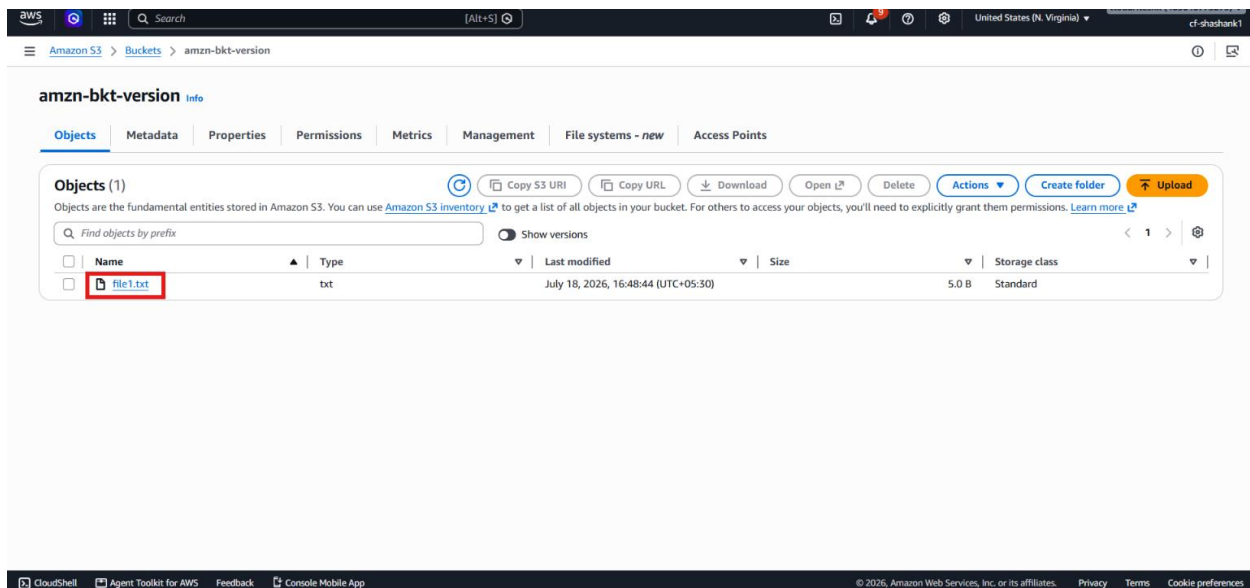
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Find objects by prefix

☒ Show versions

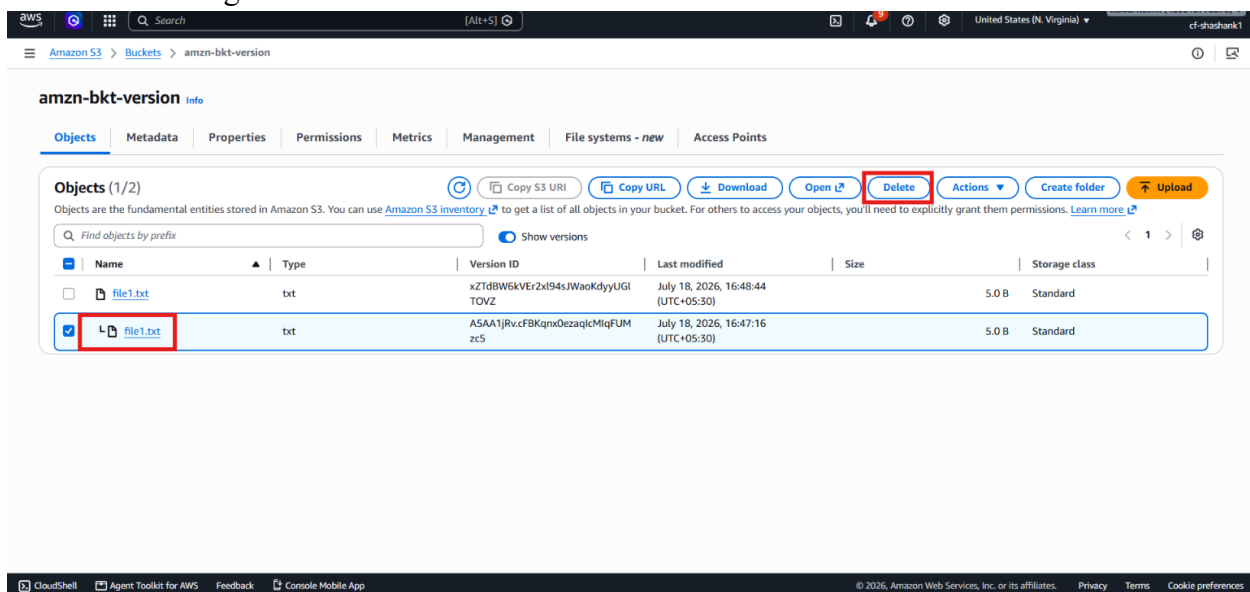
	Name	Type	Version ID	Last modified	Size	Storage class
☐	file1.txt	txt	xZTdBW6kVEr2xI94sJWaoKdyYUGITOVZ	July 18, 2026, 16:48:44 (UTC+05:30)	5.0 B	Standard
☐	file1.txt	txt	ASAA1jRvcFBKqnx0ezaqlcMiqFUMzc5	July 18, 2026, 16:47:16 (UTC+05:30)	5.0 B	Standard

- Confirm that the original file is visible again.

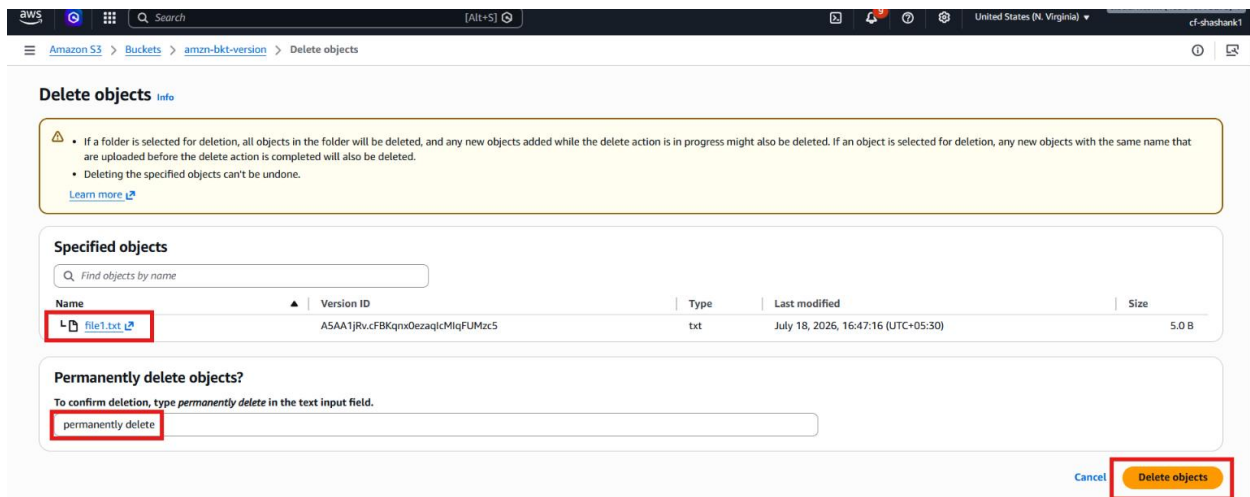


Step 9 – Permanently delete a specific version

- Open the version list.
- Select the original version.



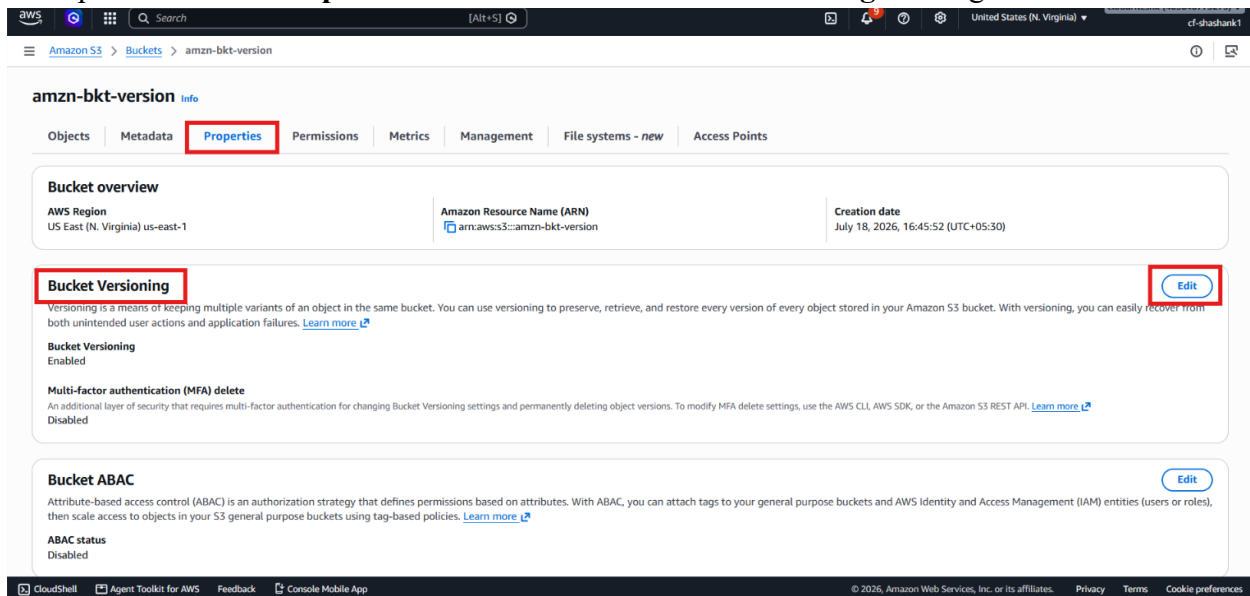
- Delete that version permanently if you want to remove the object fully.



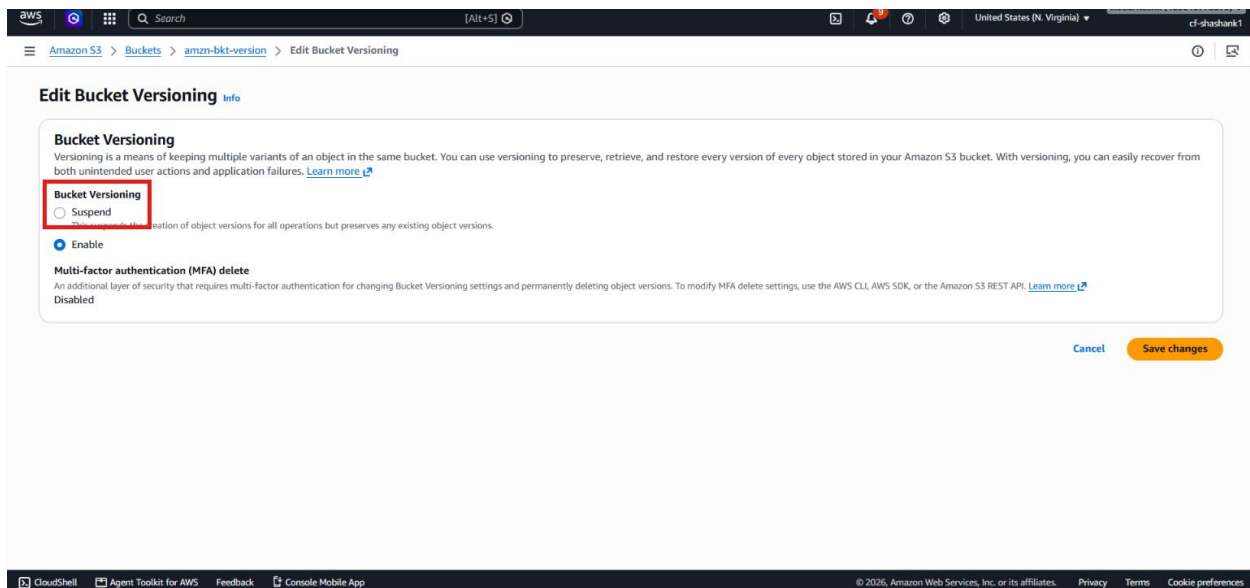
- Leave the delete marker if cleanup is not needed immediately.

Step 10 – Review versioning rules

- Open the bucket **Properties** and besides the **Bucket Versioning** heading click on **Edit**.



- Confirm that versioning can be **suspended** but not fully disabled.



- Review the impact of storage cost when multiple versions are kept.
- Note that versioning can be combined with lifecycle rules for cleanup.

Verification

- Verified that an S3 bucket with versioning disabled was created successfully.
- Verified that uploading the same file twice overwrote the old version when versioning was off.
- Verified that a second bucket with versioning enabled stored multiple object versions.
- Verified that deleting an object in a versioned bucket created a delete marker.
- Verified that the deleted file could be restored by removing the delete marker.
- Verified that versioning cannot be fully disabled once enabled and can only be suspended.
- Verified that versioned objects may increase storage cost because old versions are retained.